

Perfect-correction Markov-channel and continuous Lindblad dynamics at a class-A domain wall

Internal campaign summary¹

¹Class-A fermionic adaptive-circuit project

(Dated: Working-note scaffold; numerical claims intentionally withheld)

We define a matched comparison between the deterministic trajectory-averaged perfect-correction Markov channel and an infinitesimal continuous Lindblad evolution built from identical overcomplete Wannier modes. Both domain-wall support truncation settings and two finite spatial shells are included. The physical channel retains its number-measurement dephasing; the continuous evolution is evaluated with and without the corresponding number-dephasing term. Because a random ordered sweep need not be translation invariant along the wall, its momentum-resolved estimator is defined by a late-time temporal average followed by an exact translation-group twirl. This note fixes the equations, estimators, data provenance, and interpretation rules before any new result is inserted. The earlier gain/loss-only calculation is retained as a separately identified stationary baseline, not as the requested Markov channel.

I. CORRECTION AND SCOPE

The computed objects must be distinguished at the outset. The first is the ordered one-cycle correlation-matrix map produced by `classA.U1FGTN.run_markov_channel`. It is the deterministic average of the projective measurement and perfect-feedback protocol, not a sampled trajectory. The second is the continuous equation obtained from infinitesimal local channel increments. The third is the older auxiliary Gaussian bath calculation. At ancillary filling $\bar{n}_a = 1/2$, its dephasing-free generator is exactly one half of the perfect-correction dephasing-free continuous generator. Its stationary correlation matrix is therefore reusable, whereas its relaxation clock, response fits, and finite Strang map are not.

No result below is called a trajectory Lyapunov spectrum, a Hamiltonian band, or a Liouvillian many-body spectrum. Natural occupations are eigenvalues of a two-point correlation matrix. A midpoint crossing can diagnose a state-level purity gap without implying a vanishing relaxation rate.

II. COMMON OW GEOMETRY

We follow the notation of Bhuiyan, Pan, and Jian [1]. Many-body operators carry hats, $\mu = 1, 2$ labels physical orbitals, $\nu = A, B$ labels trial-orbital families, and $\pm$ labels upper and lower parent bands. The parent single-particle matrix in the archived run convention is

$$h_{\alpha_{\text{run}}}(\mathbf{k}) = \hat{\mathbf{n}}_{\alpha_{\text{run}}}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \quad (1)$$

$$\mathbf{n}_{\alpha_{\text{run}}}(\mathbf{k}) = (\sin k_x, \sin k_y, \alpha_{\text{run}} - \cos k_x - \cos k_y), \quad (2)$$

with $\alpha_{\text{PRR}} = -\alpha_{\text{run}}$. It is used to construct the *overcomplete Wannier (OW) basis*; a spatial domain-wall Hamiltonian is not diagonalized.

For center $\mathbf{r}$ the normalized state used by both dynam-

$$|\widetilde{W}_{\mathbf{r},\nu,\pm}\rangle = \frac{M_{n_{\text{shell}}}(\mathbf{r}' - \mathbf{r})M_{\text{DW}}(\mathbf{r}'_x; \mathbf{r}_x)|W_{\mathbf{r},\nu,\pm}\rangle}{\|M_{n_{\text{shell}}}M_{\text{DW}}|W_{\mathbf{r},\nu,\pm}\rangle\|}. \quad (3)$$

The shell mask depends on relative periodic displacement. The optional hard-wall mask allows support only in the domain containing the center. All cases use $N_x = 20$, walls $x = 5, 15$, $\alpha_{\text{run,out}} = 30$, trial basis X , and a maximally mixed initial state. These explicit walls override incompatible historical defaults.

Define

$$\hat{\chi}_{\mathbf{r},\nu,\pm} = \sum_{\mathbf{r}',\mu} \widetilde{W}_{\mathbf{r},\nu,\pm}(\mathbf{r}', \mu) \hat{c}_{\mathbf{r}',\mu}, \quad (4)$$

$$\hat{N}_{\mathbf{r},\nu,\pm} = \hat{\chi}_{\mathbf{r},\nu,\pm}^\dagger \hat{\chi}_{\mathbf{r},\nu,\pm}, \quad (5)$$

and the rank-one single-particle matrix $P_{\mathbf{r},\nu,\pm} = |\widetilde{W}_{\mathbf{r},\nu,\pm}\rangle\langle\widetilde{W}_{\mathbf{r},\nu,\pm}|$. Distinct OW stabilizers need not commute.

III. THE TWO REQUESTED DYNAMICS

A. Ordered perfect-correction channel

The published correlation convention is

$$G_{ij} = \text{Tr}(\hat{\rho} \hat{c}_i^\dagger \hat{c}_j). \quad (6)$$

The canonical class returns $Q = 2G - \mathbf{1}$; the runner records this bridge and stores physical G . For a single mode P , let $\mathcal{D}^\dagger[\hat{L}]$ denote the adjoint Lindblad action on G . The dephasing-free loss and gain terms are

$$\mathcal{D}^\dagger[\hat{\chi}](G) = -\frac{1}{2}\{P, G\}, \quad (7)$$

$$\mathcal{D}^\dagger[\hat{\chi}^\dagger](G) = P - \frac{1}{2}\{P, G\}. \quad (8)$$

Thus $\mathcal{I} + \mathcal{D}^\dagger[\hat{\chi}]$ and $\mathcal{I} + \mathcal{D}^\dagger[\hat{\chi}^\dagger]$ give the expressions sometimes called a “no-dephasing unit-step channel.” They are formal forward-Euler increments, not physical local maps for overlapping OW modes at unit step.

Number measurement contributes $\mathcal{D}^\dagger[\hat{N}](G) = PGP - \frac{1}{2}\{P, G\}$. The exact empty and filled reset maps are

$$\mathcal{R}_{P,+}(G) = G - \{P, G\} + PGP, \quad (9)$$

$$\mathcal{R}_{P,-}(G) = G + P - \{P, G\} + PGP, \quad (10)$$

Writing $\mathcal{I}$ for the identity superoperator, these maps obey

$$\mathcal{R}_{P,+} - \mathcal{I} = \mathcal{D}^\dagger[\hat{\chi}] + \mathcal{D}^\dagger[\hat{N}], \quad (11)$$

$$\mathcal{R}_{P,-} - \mathcal{I} = \mathcal{D}^\dagger[\hat{\chi}^\dagger] + \mathcal{D}^\dagger[\hat{N}]. \quad (12)$$

Equivalently, each reset is exactly $\mathcal{R}_a = \mathcal{I} + \mathcal{L}_a$, with $\mathcal{L}_a$ given by the appropriate right-hand side of Eq. (12). At each visited unit cell the canonical order is $A+$ (loss), $A-$ (gain), $B+$ (loss), $B-$ (gain); cells are visited in the saved random order. The full sweep is the corresponding noncommuting product

$$\mathcal{E}_s = \prod_{\mathbf{r} \in \mathbf{s}}^{\leftarrow} \mathcal{R}_{P_{\mathbf{r},B,-},-} \mathcal{R}_{P_{\mathbf{r},B,+},+} \mathcal{R}_{P_{\mathbf{r},A,-},-} \mathcal{R}_{P_{\mathbf{r},A,+},+}, \quad (13)$$

with schedule record $\mathbf{s}$. Equation (13) is the production channel. It is also exact to write

$$\mathcal{E}_s = \mathcal{I} + \mathcal{L}_{\text{cycle},\mathbf{s}}, \quad \mathcal{L}_{\text{cycle},\mathbf{s}} = \mathcal{E}_s - \mathcal{I}. \quad (14)$$

Here $\mathcal{L}_{\text{cycle},\mathbf{s}}$ contains every schedule-ordered cross term generated by the product; it is not $\sum_a \mathcal{L}_a$. The literal no-dephasing unit-step Euler product is excluded because it is not positivity preserving for overlapping OW modes; it is retained only as a labelled small-system diagnostic.

B. Infinitesimal continuous equation

Let $a = (\mathbf{r}, \nu, n)$, $P_a = P_{\mathbf{r},\nu,n}$, $s_a = 1$ for $n = -$, and $s_a = 0$ for $n = +$. The continuous perfect-correction action matched to the local increments above is

$$\begin{aligned} \frac{dG}{dt} &= \mathcal{L}_{\text{PC}}^{(\zeta)}(G) \\ &= \sum_a \left[s_a P_a - \frac{1}{2} \{P_a, G\} + \zeta P_a G P_a - \frac{\zeta}{2} \{P_a, G\} \right]. \end{aligned} \quad (15)$$

The $\zeta = 0$ arm contains unit gain and loss. The $\zeta = 1$ arm additionally contains unit number dephasing, equivalent at the many-body level to adding $\sum_{\mathbf{r},\nu,n} \mathcal{D}[\hat{N}_{\mathbf{r},\nu,n}]$. Although number dephasing need not preserve Gaussian many-body states, Eq. (15) closes exactly on G . Consequently an entropy evaluated only from the natural occupations is called a Gaussian entropy proxy whenever $\zeta = 1$. Equation (15) is the infinitesimal Poisson generator $\sum_a \mathcal{L}_a$ (or its dephasing-deleted version), not the finite-sweep $\mathcal{L}_{\text{cycle},\mathbf{s}}$ of Eq. (14).

The production integrator is complex128 RK4 with $\delta t = 0.05$. A separate representative validation subset at $\delta t = 0.1, 0.05, 0.025$ forms the step-convergence gate; these are not additional production physics cases. The file `CI.Lindblad.DW.py` is an algebraic reference only; production consumes the same OW vectors as the channel and does not inherit that file's distinct wall rule.

IV. ORDER, TRANSLATION, AND MOMENTUM ESTIMATORS

Because Eq. (3) is translated identically at every y and the wall mask depends only on x ,

$$[T_y, \mathcal{L}_{\text{PC}}^{(\zeta)}] = 0. \quad (16)$$

A translation-invariant initial state remains invariant under the continuous equation, and $G(k_y)$ has exact $2N_x \times 2N_x$ blocks. A finite ordered product generally does not commute with T_y , even if every local family is covariant. We therefore do not diagonalize the diagonal Fourier slices of the untwirled channel state and call them bands.

For each independent schedule sample s , first form the approved late-cycle average

$$\bar{G}_s = \frac{1}{N_y} \sum_{c=N_y+1}^{2N_y} G_s(c), \quad (17)$$

then the exact spatial sliding estimator

$$\mathcal{T}_y[\bar{G}_s] = \frac{1}{N_y} \sum_{q=0}^{N_y-1} T_y^q \bar{G}_s T_y^{-q}. \quad (18)$$

Its blocks define the channel's explicitly *twirled* spectrum. Every such plot is accompanied by the untwirled residual

$$\varepsilon_T = \frac{\|T_y \bar{G}_s T_y^{-1} - \bar{G}_s\|_F}{\|\bar{G}_s\|_F}. \quad (19)$$

For a complementary representation that makes no symmetry assumption, a global mode $G|\phi_j\rangle = \nu_j|\phi_j\rangle$ carries momentum weight

$$Z_j(k_y) = \sum_{x,\mu} |\langle x, k_y, \mu | \phi_j \rangle|^2, \quad \sum_{k_y} Z_j(k_y) = 1. \quad (20)$$

V. CAMPAIGN AND STORAGE DESIGN

Cheap observables are retained for sample s at every integer cycle $0, 1, \dots, 2N_y$. Spectral estimators are evaluated only at the explicit checkpoint cycles declared in the immutable configuration; a dense 2560×2560 matrix is not diagonalized after every cycle. Dense states are checkpoint-resolved, with final and late-average matrices retaining the sample axis. Each atomic case directory contains an NPZ product, JSON metadata, a terminal marker, and checksums. No pre-existing run directory is overwritten.

VI. FIGURE ESTIMATORS

Let

$$G(k_y)|\phi_j(k_y)\rangle = \nu_j(k_y)|\phi_j(k_y)\rangle \quad (21)$$

TABLE I. Locked production design. Alpha scans use DW truncation on only; endpoint size scans use it on and off. The independent unit is the schedule sample for the ordered channel and the deterministic run for the continuous arms.

Endpoint grid	$N_y = 32, 48, 64$; $\alpha_{\text{run}, \text{in}} = 1$; $n_{\text{shell}} = 1, 2$; truncation on/off; channel with dephasing; continuous evolution without/with dephasing.
Alpha grid	$N_y = 64$; $\alpha_{\text{run}, \text{in}} = 1, 1.5, 1.75, 1.875, 2.125, 2.25, 2.5, 3$; $n_{\text{shell}} = 1, 2$; truncation on only.
Full-frame controls	Continuous dynamics only; endpoint truncation on/off and static alpha scan with truncation on.
Timing	Main evolution through $2N_y$; late average over $N_y + 1, \dots, 2N_y$; response through $N_y/2$.
Response subset	Exactly 22 endpoint cases: finite $n_{\text{shell}} = 1$ at all sizes, both truncation settings, and all three arms; plus full-frame continuous cases at $N_y = 64$ for both truncation settings. Alpha scans, $n_{\text{shell}} = 2$, and the $S = 10$ schedule control omit response.
Random schedules	Root seed 20260814; one matched seed in the main grid; a separate canonical $N_y = 64, n_{\text{shell}} = 1$, truncated channel case with $S = 10$ independent schedule seeds.
Case count	105 unique case specifications: 27 channel, 39 continuous without dephasing, and 39 continuous with dephasing.

denote either an exact continuous block or an explicitly twirled channel block. With $\mathcal{W}_a = \{x_a - 1, x_a, x_a + 1\}$,

$$W_j^{(a)}(k_y) = \sum_{x \in \mathcal{W}_a, \mu} |\phi_{j, x\mu}(k_y)|^2. \quad (22)$$

Eligible wall modes have $W_j^{(a)} \geq 0.25$ and minimize $|\nu_j - 1/2| + 0.02(1 - W_j^{(a)})$ independently at each momentum. The displayed slope is an ordinary least-squares fit over $|k_y| \leq \max(2.5, 2\pi/N_y, 0.22)$. It is not overlap tracking or a topological index.

The global midpoint gap, bulk-excluded gap, Gaussian entropy proxy, Gaussian charge-variance proxy, and midgap wall localization are

$$\Delta_{1/2} = \min_j |\nu_j - \tfrac{1}{2}|, \quad (23)$$

$$\Delta_{\text{bulk}} = \min_{j: W_j^{\text{W}} < 0.25} |\nu_j - \tfrac{1}{2}|, \quad (24)$$

$$S_G = - \sum_j [\nu_j \ln \nu_j + (1 - \nu_j) \ln(1 - \nu_j)], \quad (25)$$

$$V_G = \text{tr}[G(\mathbf{1} - G)] = \sum_j \nu_j(1 - \nu_j), \quad (26)$$

$$\mathcal{L}_{\text{mid}} = \sum_{x \in \mathcal{W}} p_x, \quad (27)$$

where p_x is averaged over the $2N_y$ modes closest to half occupation and normalized to unit total weight. For a non-Gaussian full Fock-space state, V_G is not the true charge variance; it is explicitly reported only as a two-point Gaussian proxy. No eigenvalue outside $[0, 1]$ is clipped to manufacture an entropy; its physicality violation is reported instead.

VII. RELAXATION AND DENSITY-KICK RESPONSE

The affine continuous equation has a homogeneous covariance generator $\mathbb{L}^{(\zeta)}$; the ordered sweep has a homogeneous map $\mathbb{M}_{\text{s}}$. Their comparable relaxation scales

are

$$\gamma_{\text{cont}} = \min_{\ell \neq 0} [-\text{Re } \ell], \quad (28)$$

$$\gamma_{\text{ch}} = \min_{0 < |m| < 1} [-\ln |m|], \quad (29)$$

with exact neutral sectors removed by a stated tolerance. Complex and nonnormal spectra are retained rather than replaced by the older $2d_{\text{min}}$ formula.

A source at wall a and coordinate y_0 is kicked by

$$\hat{U}_\epsilon = \exp[-i\epsilon \hat{n}_{a, y_0}], \quad \delta G(0) = i[P_{a, y_0}, G_{\text{bg}}]. \quad (30)$$

The same homogeneous channel or generator then propagates the tangent. The wall profile and positive-lobe center are

$$\chi_a^{\text{wall}}(r, t) = \sum_{x \in \mathcal{W}_a, \mu} [\delta G(t)]_{x, y_0 + r, \mu; x, y_0 + r, \mu}, \quad (31)$$

$$y_{+, a}(t) = \frac{\sum_r \tilde{r} [\chi_a^{\text{wall}}(r, t)]_+}{\sum_r [\chi_a^{\text{wall}}(r, t)]_+}, \quad (32)$$

where $\tilde{r} = ((r + N_y/2) \bmod N_y) - N_y/2$. Velocity is the OLS slope over $2 \leq t \leq 0.375N_y$, subject to the saved response-norm threshold. For the ordered channel, all y_0 are retained before averaging, so the schedule-origin spread is a deterministic systematic rather than a statistical confidence interval.

The size and direct-dynamics checks in Fig. 4 replace the old auxiliary finite- p Strang comparison. The physical channel has one unit reset per stabilizer, so no weak-step order is inferred from it.

VIII. VALIDATION AND INTERPRETATION RULES

Production launch remains gated until the following checks pass: OW projectors agree between the two engines; the local equations agree with the canonical channel on small systems; the continuous action agrees with the derivative of weakened local maps; RK4 shows the expected step convergence; the full-dephasing channel

Corrected production figure will be inserted only after manifest completion.

FIG. 1. Planned canonical spectra. All natural occupations are shown for the full-dephasing channel, continuous no-dephasing evolution, and continuous full-dephasing evolution. Channel panels are explicitly marked as spectra of Eq. (18); continuous panels use exact momentum blocks. Each caption will report initialization, evolution window, independent sample count, and translation residual.

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FIG. 2. Planned matched alpha scan. Columns report the global and bulk-excluded midpoint gaps, S_G/N_y , midgap wall localization, and relaxation. The vertical line at $\alpha_{\text{run},\text{in}} = 2$ is a parent-model reference, not a measured campaign point. Curves will distinguish shell, dynamics, and dephasing without mixing the truncation-off endpoint control into the static scan.

remains physical; continuous translation residuals meet tolerance; cycle coordinates and sample IDs are complete; the response tangent agrees with a finite central difference; and all hashes survive analysis unchanged.

The completed note may report stationary or late-state occupation flow, relaxation, and damped directional response. It may not infer a trajectory Lyapunov boundary mode, quantized transport, or schedule-independent chi-

rality. The random ordered channel's translation twirl is an estimator applied to its late state, not a symmetry retroactively assigned to the underlying map.

This source is maintained under a strict nine-page main-text ceiling after figures and tables are populated. Detailed schemas, derivations, convergence tables, and legacy reconciliation belong in the companion supplement and analysis notebook.

[1] A. Bhuiyan, G. Pan, and S.-K. Jian, “Free-fermion dynamics with measurements: Topological classification and

adaptive preparation of topological states,” *Phys. Rev. Research* **8**, 023147 (2026), doi:10.1103/nk38-ygyq.

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FIG. 3. Planned local density-phase response. Separate rows compare the channel and two continuous arms; columns correspond to the two walls. The channel profile is source-resolved in the saved product and source-averaged only for display. Fit points, fit window, wall retention, and deterministic R^2 will be stated.

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FIG. 4. Planned endpoint checks. Left: raw fitted velocities versus $1/N_y$ for both walls and both DW-support settings. Right: finite unit-step channel minus the dephasing-matched continuous result, including state, occupation, response-profile, and velocity differences. No thermodynamic extrapolation or weak-step convergence fit is performed.